

Module 1: Consigned Assembly & Tiered Scaling Strategy

1. Production Architecture Overview

To deliver 10,000 folding electric motorcycles in a single year, production is structured under an asset-light Consigned Assembly model. By procuring components directly at scale, the business eliminates a standard 10% to 15% manufacturer material markup, paying strictly for labor, machine utilization, and factory overhead.

The production cadence requires an output of approximately 40 completed motorcycles per working day based on a 250-day operational calendar. This equates to a manufacturing throughput velocity of 2.5 units per hour across two concurrent 8-hour shifts.

2. Tiered Production Decoupling

Manufacturing is decoupled into separate tiers to leverage specialized capabilities, optimize quality control, and prevent capital bottlenecks under a single factory roof:

- **Tier 2 Structural Frame Fabrication:** Specialized mobility and automotive fabricators handle precision cutting, bending, and robotic welding of the high-tensile **6061-T6 aluminum** structural frame parts. Frame components are heat-treated to T6 temper specifications post-welding to maximize fatigue life at the folding mechanism interfaces.
- **Tier 2 Powertrain Integration:** Dedicated electronics assembly houses construct and isolate the high-voltage electrical subsystem. This includes fabricating the modular battery tray (16S2P NMC configuration layout), routing heavy-gauge wiring harnesses, and configuring the high-amp motor controller units.
- **Tier 1 Final Assembly Hub:** All certified sub-assemblies and structural **6061-T6 aluminum** frame elements converge at a central Tier 1 mobility facility. The sole mandate of this hub is vehicle 'marriage'—dropping the integrated powertrain into the **6061-T6 aluminum** frame, mounting body plastics, performing cosmetic detailing, and packing units into transport crates.

3. Operational Logistics and Staging

A rolling 90-day inventory buffer of scaled components is maintained at the Tier 1 hub to mitigate global transit and customs delays. Component arrivals are strictly synchronized with the Tier 1 facility's monthly assembly schedule, utilizing Just-In-Time (JIT) delivery loops from the Tier 2 sub-assemblers to maximize floor space efficiency.